

Statistical Learning Project 2024/2025

Aircraft Price Analysis

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Abstract

We analyze a dataset of 507 commercial aircraft to predict market prices using a range of statistical learning techniques, including linear regression models, generalized additive models (GAM), and tree-based ensemble methods such as random forests and bagging. The study explores both predictive accuracy and model interpretability, with a focus on identifying the most influential aircraft features and evaluating the impact of different modeling strategies. Particular attention is given to the trade-offs between accuracy and explainability, which are critical in high-stakes valuation contexts. Our findings offer practical guidance on model selection for real-world aircraft pricing applications, balancing statistical performance with transparency and usability.

1 Description of the problem

1.1 Global aircraft market outlook

The global commercial aircraft market, valued at USD 426.42 billion in 2024 and projected to reach approximately USD 601.51 billion by 2034, represents one of the most complex and dynamic sectors in global manufacturing. This exponential growth trajectory underscores the critical importance of accurate valuation methodologies in an industry where individual units can range from hundreds of thousands to hundreds of millions of dollars.

1.2 Limitations of Traditional Valuation

Traditional aircraft valuation methods, rooted in expert assessment and rule-of-thumb calculations, are increasingly inadequate for modern market demands. New technologies such as electric engines, ad-

vanced composite materials, and integrated avionics systems render historical valuation models obsolete. Moreover, regulatory uncertainty, supply chain volatility, and rapidly shifting market conditions introduce layers of complexity that exceed the capacity of conventional approaches.

The industry desperately needs fast, data-driven models capable of providing real-time, objective pricing estimates. As noted by Aviation Week (2025), "Having a reliable data-based pricing model would be a game changer" for an industry where pricing decisions directly impact multi-billion dollar transactions.

2 Description of the dataset

2.1 Dataset characteristics

We utilize a comprehensive aircraft dataset sourced from Kaggle's Aircraft Price Analysis & Prediction Dataset, comprising 507 initial observations of commercial aircraft. After data cleaning procedures (detailed in Section 4.1), our final analytical dataset contains 497 observations across 14 variables: 12 numerical features capturing technical specifications and performance metrics, and 2 categorical variables representing model classifications.

The dataset encompasses a diverse range of aircraft types, from small regional jets to large commercial airliners, providing com-

prehensive coverage of the commercial aviation market. The response variable is sale price in USD, exhibiting substantial variability from \$50,000 to \$15 million, with pronounced right-skewness (skewness coefficient = 2.84).

2.2 Numerical variable

The dataset includes the following variables:

- **engine_power:** Engine power output measured in horsepower (HP), representing the fundamental propulsive capability
- **max_speed:** Maximum achievable speed in nautical miles per hour (knots), indicating peak performance capability
- **cruise_speed:** Optimal cruising speed in knots, representing operational efficiency at standard flight conditions
- **stall_speed:** Minimum speed required to prevent aerodynamic stalling in knots, a critical safety parameter
- **fuel_tank:** Total fuel tank capacity in liters, determining operational range and payload trade-offs
- **range:** Maximum operational range in nautical miles under standard conditions

- `length`: Overall aircraft length in inches, indicating size category and passenger/cargo capacity
- `wing_span`: Total wingspan in inches, affecting aerodynamic efficiency and airport compatibility
- `empty_weight`: Aircraft weight in pounds when fully equipped but without fuel, passengers, or cargo
- `all_eng_roc`: Rate of climb with all engines operational (feet per minute), indicating performance capability
- `one_eng_roc`: Single-engine rate of climb (feet per minute), a critical safety performance metric
- `takeoff_distance`: Required runway length for takeoff under standard conditions (feet)
- `landing_distance`: Required runway length for landing under standard conditions (feet)

2.3 Categorical Variables

Two categorical variables provide model classification:

- `model_name`: Specific aircraft model designation (e.g., Boeing 737, Airbus A320)
- `engine_type`: Engine technology classification (turbofan, turboprop, etc.)

3 Description of the Scientific Questions

Our research addresses five primary scientific questions:

Q1: Model Performance Comparison - Which statistical learning methods provide the most accurate aircraft price predictions, and what is the magnitude of performance differences between linear, regularized, and ensemble approaches?

Q2: Feature Importance and Value Drivers - Which aircraft characteristics are the most important predictors of market value, and how do feature importance rankings vary across different modeling approaches?

Q3: Nonlinear Relationships and Interactions - Do aircraft pricing relationships exhibit significant nonlinearity and feature interactions that justify the use of advanced ensemble methods over traditional linear approaches?

Q4: Practical Implementation - Which models provide the optimal balance of accuracy, interpretability, and computational efficiency for real-world aircraft valuation applications?

Hypotheses: Based on domain knowledge, we hypothesize that: (1) Tree-based ensemble methods will outperform linear approaches due to complex feature interactions, (2) Aircraft range and speed characteristics will emerge as primary value drivers, (4) Random Forest will provide optimal accuracy-interpretability balance.

4 Data analysis

4.1 Data Preprocessing

4.1.1 Data Cleaning and Missing Value Treatment

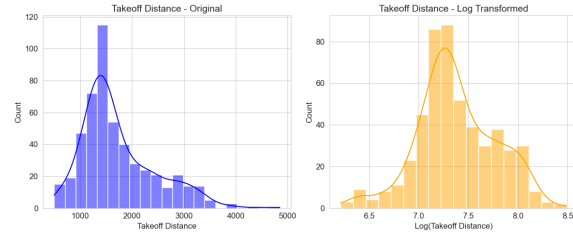
Our preprocessing pipeline begins with comprehensive data quality assessment. The original dataset contained 507 observations, with 10 cases exhibiting missing values in the target variable (`price`). We removed these incomplete cases, yielding a final analytical dataset of 497 observations.

4.1.2 Logarithmic Transformation

Aircraft specifications exhibit heavy-tailed, right-skewed distributions that violate linear regression assumptions. We apply logarithmic transformation to all numerical variables:

$$x' = \ln(x)$$

This transformation provides multiple benefits: symmetry enhancement, variance stabilization, linearity promotion and outlier mitigation. Figure below demonstrates the transformation's effectiveness in normalizing distributions and reducing skewness coefficients from 2-4 to approximately 0-1.



Feature distributions before and after logarithmic transformation.

4.2 Linear regression

We begin with ordinary least squares (OLS) regression as our baseline approach. Linear regression assumes a linear relationship between predictors and response:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_p x_p + \epsilon$$

where $\epsilon \sim N(0, \sigma^2)$ represents the error term. The coefficients are estimated by minimizing the residual sum of squares:

$$\hat{\beta} = \arg \min_{\beta} \|y - X\beta\|_2^2 = (X^T X)^{-1} X^T y$$

For the whole project we evaluate model performance primarily using Root Mean Squared Error (RMSE), which measures the average prediction error in USD, and the coefficient of determination (R^2), which quantifies the proportion of variance in price explained by the model. Lower RMSE and higher R^2 indicate better predictive accuracy.

Using the full model with all 13 numerical features, we first fit using 10-fold cross-validation with shuffling. Initial results yielded:

- **R²**: 0.8282
- **MSE**: 0.0275
- **RMSE (USD)**: \$391,503

Feature	β_{mean}	p _{mean}
engine_power	0.011	0.543
max_speed	0.175	0.001
cruise_speed	0.504	0.001
stall_speed	0.001	0.430
fuel_tank	0.039	0.038
all_eng_roc	0.119	0.001
out_eng_roc	-0.099	0.005
takeoff_distance	0.121	0.005
landing_distance	-0.325	0.001
empty_weight	0.104	0.008
length	0.155	0.178
wing_span	0.367	0.001
range	0.025	0.336

Analysis of coefficient statistical significance revealed several features with p-values exceeding 0.05: `engine_power` (p = 0.543), `stall_speed` (p = 0.430), `length` (p = 0.178), and `range` (p = 0.336). We constructed a reduced model excluding these non-significant features, retaining 9 statistically significant variables (p < 0.05).

The reduced OLS model performance showed marginal improvement:

- **R²**: 0.8291
- **MSE**: 0.0271
- **RMSE (USD)**: \$389,000

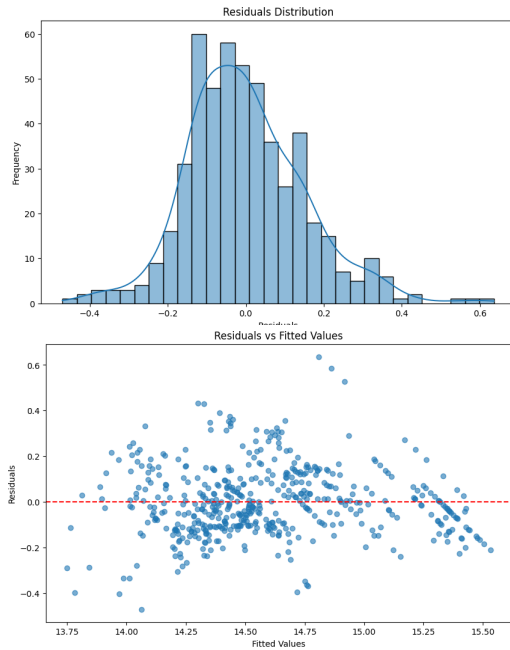
Feature	β_{mean}	p _{mean}
intercept	8.515	0.001
max_speed	0.179	0.001
cruise_speed	0.522	0.001
fuel_tank	0.049	0.001
all_eng_roc	0.125	0.001
out_eng_roc	-0.084	0.013
takeoff_distance	0.132	0.001
landing_distance	-0.285	0.001
empty_weight	0.109	0.006
wing_span	0.405	0.001

This modest improvement suggests that the excluded features contribute primarily noise rather than signal, validating our feature selection approach.

4.2.1 Residual analysis

Residuals exhibit a symmetric distribution centered around zero, with no discernible patterns or systematic structures visible in the residual plot. This indicates that the model effectively captures the underlying relationships between the predictor variables and the response. Moreover, the absence of heteroscedasticity suggests constant variance of errors, satisfying a key assumption of ordinary least squares regression. Although a few observations show larger residual values, these points do not qualify as extreme outliers and likely represent natural variability in the data rather than model misspecification or influential anomalies. Overall, the residual analysis supports the adequacy and reliability of the

reduced regression model for inference and prediction.



OLS; as $\lambda \rightarrow \infty$, coefficients approach zero. The Ridge solution has the closed form:

$$\hat{\beta}_{\text{ridge}} = (X^T X + \lambda I)^{-1} X^T y$$

Cross-Validation for λ Selection: We used 5-fold cross-validation to select the optimal λ value from a grid spanning $[10^{-4}, 10^4]$. The optimal value was $\lambda = 0.7009$, selected based on minimum cross-validation MSE.

Ridge Results:

- **Best α :** 0.7009
- **R^2 :** 0.8495
- **MSE:** 0.0265
- **RMSE :** \$370,000

4.3 Shrinkage Methods

Traditional OLS can suffer from high variance when p is large relative to n , or when predictors are highly correlated. Shrinkage methods address these issues by constraining or shrinking coefficient estimates toward zero.

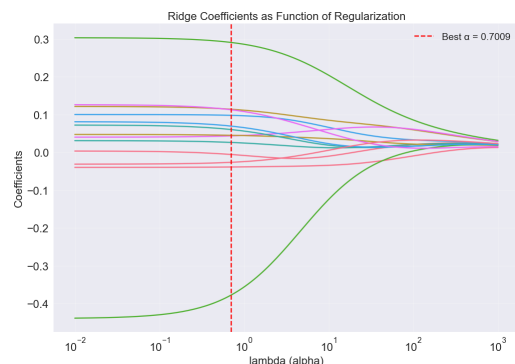
4.3.1 Ridge Regression

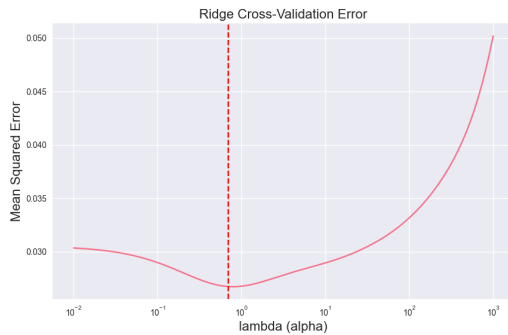
Ridge regression adds an L2 penalty term to the OLS objective function:

$$\hat{\beta}_{\text{ridge}} = \arg \min_{\beta} \{ \|y - X\beta\|_2^2 + \lambda \|\beta\|_2^2 \}$$

The penalty parameter $\lambda \geq 0$ controls the amount of shrinkage. When $\lambda = 0$, we get

Ridge regression successfully reduced overfitting while maintaining all features in the model. The coefficient shrinkage provided improved generalization performance compared to OLS.





4.3.2 Lasso Regression

Lasso (Least Absolute Shrinkage and Selection Operator) employs an L1 penalty:

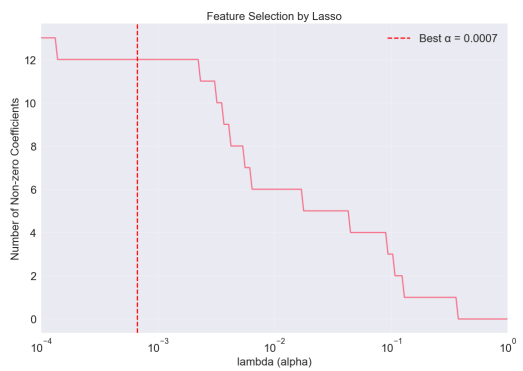
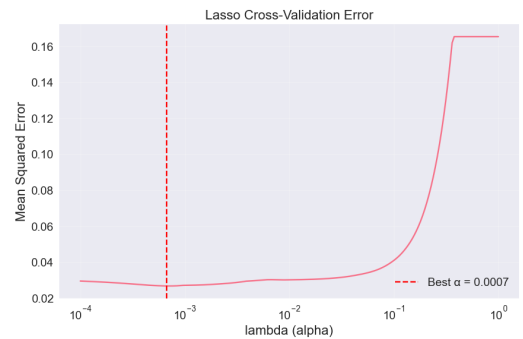
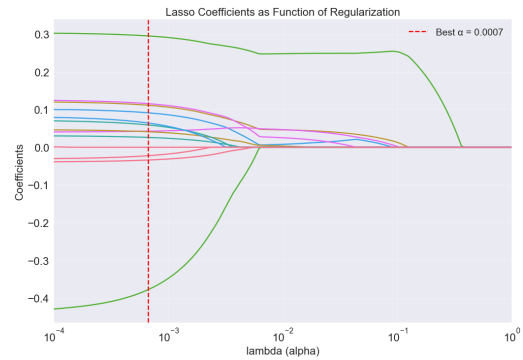
$$\hat{\beta}_{\text{lasso}} = \arg \min_{\beta} \{ \|y - X\beta\|_2^2 + \lambda \|\beta\|_1 \}$$

Unlike Ridge, Lasso can shrink coefficients exactly to zero, performing automatic feature selection. The L1 penalty creates a non-differentiable objective that requires specialized optimization algorithms like coordinate descent.

Cross-Validation for λ Selection: Optimal $\lambda = 0.0007$ was selected via 5-fold CV, automatically selecting 12 of 13 features (one coefficient driven to exactly zero).

Lasso Results:

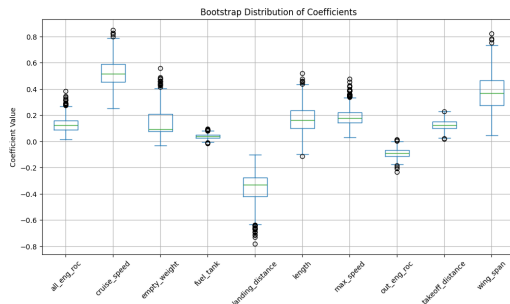
- **Best α :** 0.0007
- **R^2 :** 0.8494
- **MSE:** 0.0265
- **RMSE :** \$376,000
- **Selected features:** 12 / 13



The minimal performance difference between Ridge and Lasso suggests robust feature importance across regularization approaches, with Lasso providing the additional benefit of automatic feature selection.

4.4 Bootstrap inference

We performed 1000 bootstrap replications to assess coefficient stability without assuming specific distributions. In Lasso, frequent zero estimates reflect its feature selection capability, whereas Ridge shrinks coefficients without eliminating them. Overall, bootstrap analysis confirms the reliability and interpretability of our regularized models.



Each boxplot shows the variability of coefficient estimates across 1000 bootstrap replications. Wider intervals reflect greater uncertainty in feature influence.

5 Generalized additive models (GAM)

To capture potential nonlinear relationships between aircraft characteristics and price, we implemented Generalized Additive Models using **smoothing splines**.

5.1 GAM implementation

We built a multivariate GAM over the log-price target using all 13 numeric features:

$$\hat{y} = \beta_0 + \sum_{j=1}^{13} f_j(x_j)$$

where each f_j is a univariate smoothing spline term. The model uses:

- **Spline terms:** one spline $s(j)$ for each of the 13 numeric predictors
- **Automatic tuning:** grid-search on penalties λ_j via GCV using PyGAM's gridsearch

5.1.1 GAM performance

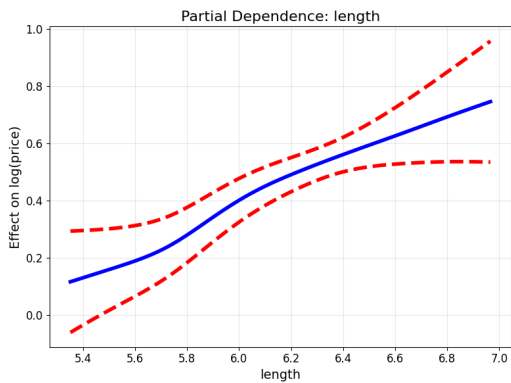
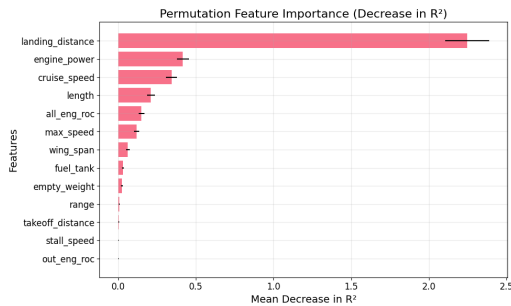
The final GAM model achieved excellent performance with good generalization:

Metric	Train	Test
RMSE	\$302,712	\$361,902
R^2	0.9038	0.8783

The small train–test gap in both RMSE and R^2 indicates good generalization. AIC (17,014.21) and GCV (0.0204) values confirm the GAM's superior fit over linear and regularized benchmarks, with an effective degrees of freedom (EDF) of 38.09.

5.1.2 Partial dependence analysis

GAM partial dependence plots uncover clear nonlinear effects: `cruise_speed` has a strong positive impact with noticeable inflection points, `engine_power` exhibits diminishing returns at higher levels, and weight-related features follow more complex, nonmonotonic patterns.



5.1.3 GAM residual analysis

Residual analysis confirms model validity with homoscedasticity and approximate normality, supporting model assumptions.

6 Tree-based methods

To explore complex feature interactions and nonlinear patterns, we implemented a comprehensive suite of tree-based methods following ISLR methodology.

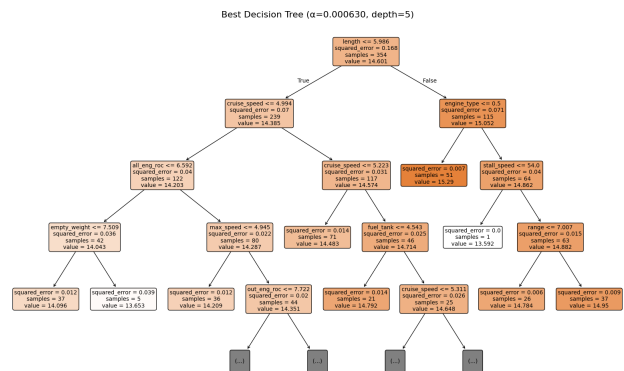
6.0.1 Decision trees with pruning

We applied cost-complexity pruning to optimize tree structure:

$$C_{\alpha}(T) = \sum_{m=1}^{|T|} N_m Q_m(T) + \alpha |T|$$

The pruning process involved growing a full tree and pruning via cost-complexity, selecting optimal α^* by 5-fold cross-validation. The optimal pruned tree achieved:

α^* (CV)	0.00063
Final depth	5
Test RMSE	\$326,069
Test R^2	0.8587

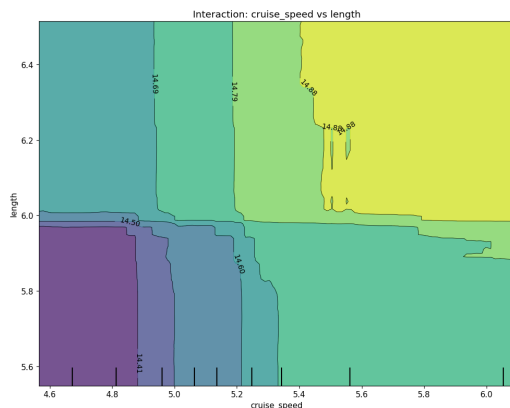


6.0.2 Ensemble methods

We implemented three ensemble approaches to reduce variance and improve prediction accuracy:

Random Forest: Optimized via grid search with 5 folds CV we achieved:

Metric	Value
Test RMSE	\$282,517
Test R^2	0.9056
OOB R^2	0.9114



Random Forest partial dependence showing interaction between cruise_speed and length, with darker regions indicating lower predicted prices and lighter (yellow) regions in the top right indicating higher predicted prices.

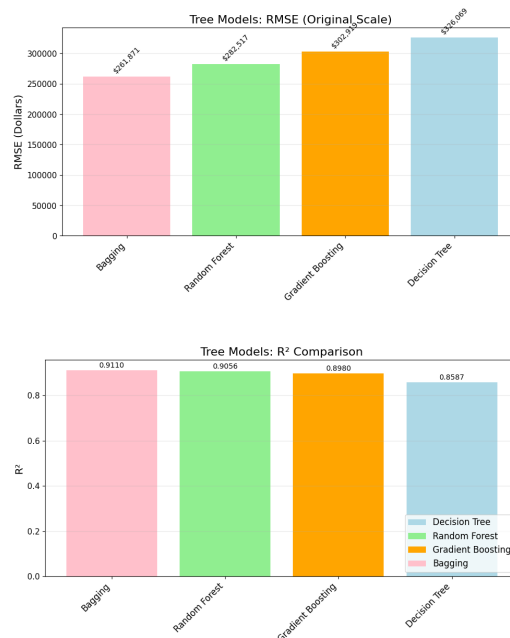
Gradient Boosting: Applied with various learning rates and tree depths:

Metric	Value
Test RMSE	\$302,919
Test R^2	0.8980
Best learning rate	0.1

Bagging (Extra Trees): Implemented using all features to represent pure bagging. This approach achieved the best overall performance:

Metric	Value
Test RMSE	\$261,871
Test R^2	0.9110

6.0.3 Tree methods comparison



6.0.4 Residual analysis for tree methods

Comprehensive residual analysis across all tree-based methods shows homoscedasticity with uniform residual spread and acceptable normality. Random Forest exhibits the lowest residual variance, indicating the best fit.

7 Overall comparison

7.1 Overall model performance

Comprehensive evaluation across all implemented methods reveals clear performance hierarchies:

Model	RMSE	R ²
Lin. Regression	\$389,179	0.8421
Ridge	\$370,000	0.8495
Lasso	\$376,000	0.8494
GAM	\$361,000	0.8920
Decision Tree	\$326,069	0.8587
Random Forest	\$282,517	0.9056
Gradient Boosting	\$302,919	0.8980
Bagging	\$261,871	0.9110

7.2 Performance by Model

Linear methods: Ridge and Lasso demonstrate improvements over basic linear regression due to regularization techniques, addressing multicollinearity effectively. However, they remain moderately accurate ($R^2 \approx 0.85$).

Nonlinear methods: GAM substantially enhances predictive performance by capturing nonlinear relationships ($R^2 = 0.8920$), thus bridging accuracy and interpretability.

Tree-based methods: Decision trees provide a straightforward interpretability baseline, but ensemble approaches (Random Forest, Gradient Boosting, Bagging) clearly

dominate, with Bagging (Extra Trees) offering optimal performance by balancing accuracy, robustness, and stability.

8 Conclusions

Our evaluation of statistical learning methods for aircraft valuation highlights distinct trade-offs between predictive accuracy and interpretability. While ensemble methods like Bagging (Extra Trees) provide superior accuracy, reducing errors substantially, their complexity makes them best suited for applications prioritizing prediction quality. Generalized additive models (GAM) effectively balance interpretability with strong predictive performance, capturing critical nonlinear relationships through intuitive visualizations. For contexts requiring clear interpretability and transparency—such as regulatory compliance or stakeholder reporting—linear regression approaches (Ridge and Lasso) remain ideal due to their simplicity and explanatory power.

In practice, the optimal method depends critically on the intended application, the necessity for interpretability, and the regulatory environment, enabling informed and effective decision-making tailored specifically to real-world aviation industry needs.

Acknowledgments and References

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Reference:

- James, G., Witten, D., Hastie, T., Tibshirani, R., & Taylor, J. (2023). *An Introduction to Statistical Learning: with Applications in Python*. Springer Cham.